

BUSINESS INTELLIGENCE REPORT

DECODING DEMAND

A Power BI Analysis of Growth, Retention & Fulfillment
Across Brazil's Olist E-Commerce Marketplace

99,441

ORDERS ANALYZED

96,096

UNIQUE CUSTOMERS

3.1%

RETENTION RATE

\$13.6M

TOTAL REVENUE

PREPARED BY

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Business Analytics | Power BI • Power Query • DAX

DATA SCOPE

Olist Public E-Commerce Dataset, Brazil
Observation Window: 2016 – 2018

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This report distills a five-page interactive Power BI dashboard, built end-to-end on the Olist public e-commerce dataset, into a decision-ready narrative for stakeholders in growth, retention, and operations strategy.

SECTION 01

Executive Summary

Olist's marketplace data reveals a business defined by strong top-line momentum and a structural weakness in customer loyalty. Between 2016 and 2018, the platform processed **99,441 orders** from **96,096 unique customers**, generating an estimated **\$13.6 million** in gross sales. Order volume grew more than **130% year-over-year** between 2017 and 2018, driven almost entirely by new-customer acquisition rather than repeat purchasing — only **3.1% of customers** ever placed a second order.

Demand is heavily concentrated in São Paulo and the broader Southeast region, which together account for the majority of both order volume and seller presence. Fulfillment performance is respectable at the aggregate level — a **12.1-day average delivery window** with a **97% successful delivery rate** — but a closer look shows delivery speed is directly tied to regional order density, exposing a fulfillment gap in lower-volume states.

“The central finding of this analysis is not that Olist is growing — it clearly is — but that its growth engine runs almost exclusively on acquisition. Retention is the single largest unrealized lever for sustainable, capital-efficient growth.”

99,441 TOTAL ORDERS Jan 2016 – Sep 2018	\$13.6M GROSS SALES Order-item value	3.1% RETENTION RATE 2,997 of 96,096 customers	12.1 days AVG. DELIVERY 8.1% delivered late
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Report Purpose

This document accompanies an interactive Power BI dashboard developed to demonstrate applied business analytics capability: relational data modeling, DAX measure design, and executive-level insight communication. It is structured to mirror the five analytical pages of the live dashboard.

SECTION 02

Methodology & Data Scope

The analysis is built on five relational tables from the Olist public dataset — orders, customers, products, sellers, and product category translations — modeled in Power BI using a star-schema approach with a dedicated date dimension table.

Data Preparation

- Standardized customer identity using **customer_unique_id** rather than per-order **customer_id**, correcting a structural risk of overcounting unique customers.
- Preserved zip codes as text to retain leading zeros lost under default numeric typing.
- Reconciled 610 products with missing category labels and 2 categories absent from the English translation table.
- Flagged and retained date-logic anomalies (carrier date preceding approval date in 1,359 rows) for transparency rather than silent deletion.

Scope & Limitations

The dataset spans September 2016 to October 2018, with statistically thin coverage outside the January 2017 – August 2018 core window; edge months are noted but not used to drive headline conclusions. No cost or margin data is available in the source tables, so profitability metrics are intentionally excluded rather than estimated speculatively.

Table	Rows	Role in Model
Orders	99,441	Fact table — transaction grain
Customers	99,441	Dimension — identity & geography
Products	32,951	Dimension — category & attributes
Sellers	3,095	Dimension — fulfillment origin
Category Translation	71	Lookup — English labels

SECTION 03 — DASHBOARD PAGE 1 & 2

Growth & Sales Performance

Order volume climbed steadily through 2017 before sustaining a materially higher plateau through mid-2018. The trajectory reflects genuine platform-level demand growth rather than a single seasonal spike.

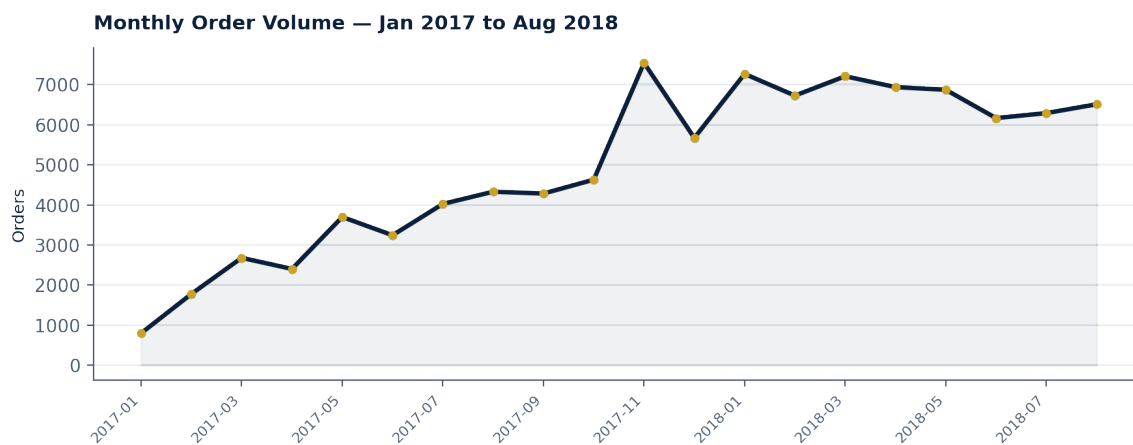


Figure. Monthly order volume, January 2017 – August 2018. November 2017 marks the single highest month at 7,544 orders.

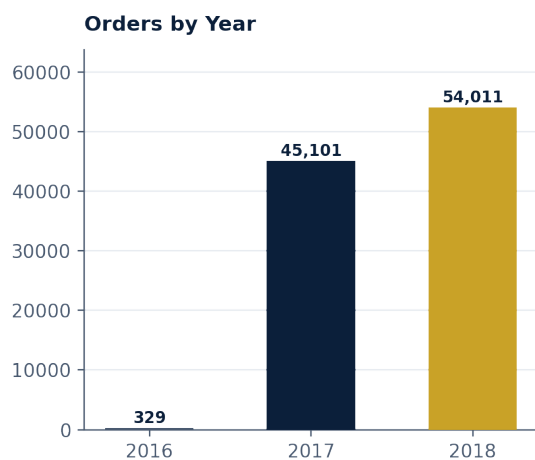


Figure. Orders by calendar year; 2018 is a partial year.

Orders grew from 45,101 in 2017 to 54,011 in 2018 (partial-year, through August) — a run-rate consistent with sustained double-digit expansion.

Comparing equivalent January–August windows, order volume rose more than **130% year-over-year**, indicating the growth curve is structural rather than a base-effect artifact of a single strong month.

SECTION 04 — DASHBOARD PAGE 3

Geographic Concentration

Demand and supply are both heavily concentrated in Brazil's Southeast. São Paulo alone represents approximately **42% of total order volume**, more than three times the next-largest state.

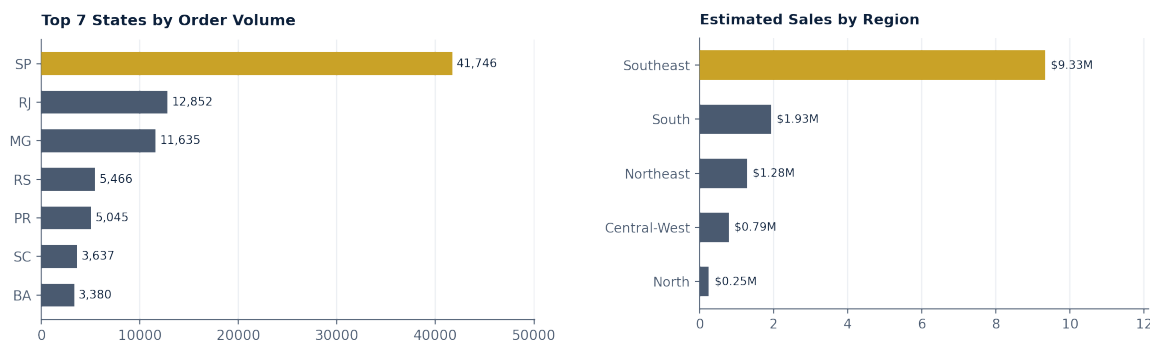


Figure. Left: order volume by top 7 states. Right: estimated sales distribution by macro-region, proportionally derived from order share.

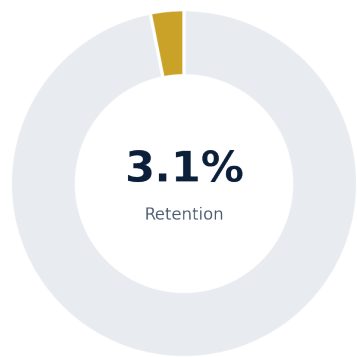
- The **Southeast region** generates roughly 4x the sales of the next-highest region, reflecting both population density and seller concentration.
- Seller presence mirrors customer demand almost exactly — 60% of active sellers are based in São Paulo state alone.
- No single customer accounts for a disproportionate share of revenue, indicating a broad, transaction-driven base with low concentration risk.

SECTION 05 — DASHBOARD PAGE 4

Customer Retention Analysis

This is the report's most consequential finding. Of 96,096 unique customers, only **2,997 (3.1%)** have ever placed a second order. Growth to date has been financed almost entirely through new-customer acquisition.

Customer Retention



96,096 total customers acquired

2,997 customers returned for a second purchase

93,099 customers never returned

At current acquisition-led economics, every incremental dollar of growth requires a new customer rather than a repeat one — a materially higher cost structure than a retention-supported growth model would require.



SECTION 06 — DASHBOARD PAGE 5

Fulfillment & Delivery Performance

Average delivery time across the platform is **12.1 days**, with **8.1% of orders** arriving after their estimated delivery date. Performance is not uniform — it is a direct function of regional order density.

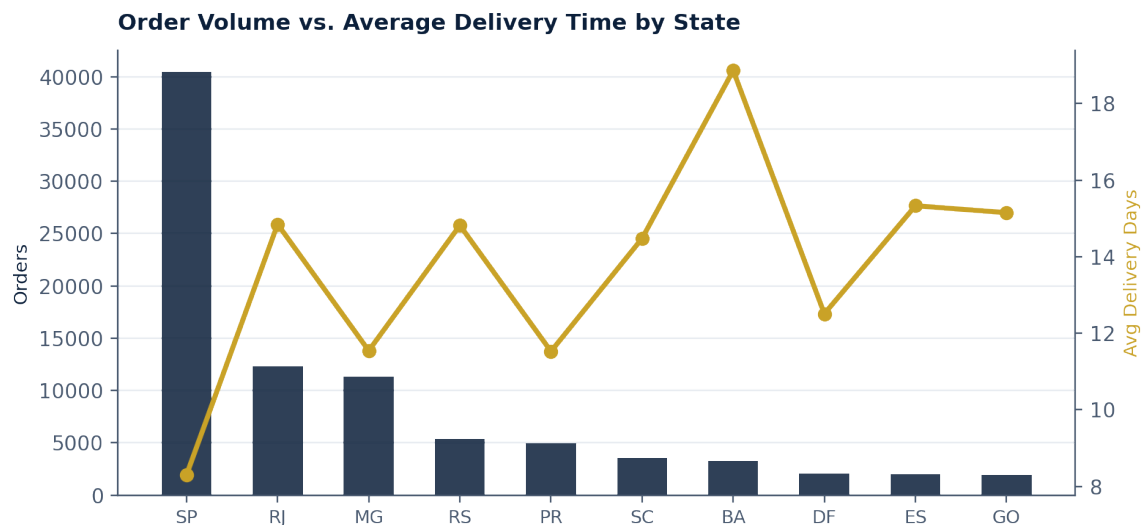


Figure. Order volume (bars) against average delivery time in days (line), by top 10 states. High-volume states consistently deliver faster.

- States with the highest order density benefit from tighter, more efficient delivery networks and shorter transit times.
- Lower-volume states show a measurable delivery-speed penalty, pointing to underinvestment in regional logistics infrastructure outside the Southeast corridor.
- A **97% overall fulfillment rate** confirms the platform's core logistics function is reliable; the opportunity lies in narrowing regional variance, not fixing a broken system.

SECTION 07

Strategic Recommendations

The following recommendations translate the analysis into prioritized, actionable initiatives, sequenced by expected impact on sustainable growth economics.

1 Launch a Structured Retention Program

Introduce post-purchase engagement triggers (targeted offers, loyalty incentives, delivery-experience follow-ups) aimed at converting first-time buyers into repeat customers, directly addressing the 3.1% retention rate.

2 Invest in Regional Fulfillment Infrastructure

Prioritize micro-warehousing or logistics partnerships in high-growth, lower-density states to close the delivery-speed gap and reduce late-delivery exposure outside the Southeast corridor.

3 Diversify Demand Beyond the Southeast

Deploy targeted marketing and seller-recruitment investment in underrepresented regions to reduce concentration risk and unlock underpenetrated demand.

4 Institute Retention as a Core Growth KPI

Elevate repeat-purchase rate to an executive-level metric alongside order volume and revenue, ensuring growth strategy accounts for acquisition cost sustainability.

SECTION 08

Conclusion & Author's Note

Olist's marketplace demonstrates strong, structurally sound top-line growth: order volume has scaled consistently, fulfillment is reliable at the aggregate level, and the platform has proven it can acquire customers at scale. The strategic gap is retention — a lever that, if addressed, would materially improve the economics of every dollar already being spent on acquisition.

This analysis was conducted independently as an applied demonstration of end-to-end business intelligence capability — from raw data modeling in Power Query, through DAX-driven measure design, to executive-level insight synthesis. The accompanying interactive dashboard is available for further exploration.

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Data Source: Olist Public E-Commerce Dataset (Brazil). All figures derived from the source tables described in Section 02; sales are order-item level values. This report and its dashboard are independent, self-directed analytical work.